

Special Relativity Notes

Invariance under Lorentz Transformations, the Lorentz Group and Minkowski Diagrams

29/09/26

1 Invariance under the Lorentz Transformation

$$x^\mu \rightarrow x'^\mu = \Lambda^\mu{}_\nu x^\nu \quad \text{so that the invariant quantity} \quad x'^\mu x'_\mu = x^\alpha x_\alpha$$

Using the Minkowski metric to lower an index, $x'^\mu x'_\mu = g^{\mu\nu} x'_\nu x'_\mu$:

$$\begin{aligned} x'^\mu x'_\mu &= g^{\mu\nu} \Lambda^\alpha{}_\nu x_\alpha \Lambda^\beta{}_\mu x_\beta \\ &= g^{\alpha\beta} x_\alpha x_\beta = x^\alpha x_\alpha \end{aligned}$$

Since this must hold for arbitrary x_α, x_β :

$$g^{\mu\nu} \Lambda^\alpha{}_\mu \Lambda^\beta{}_\nu = g^{\alpha\beta}$$

$$\begin{aligned} \Rightarrow \quad \Lambda^\alpha{}_\mu g^{\mu\nu} \Lambda^\beta{}_\nu &= g^{\alpha\beta} \\ \Rightarrow \quad \Lambda^\alpha{}_\mu g^{\mu\nu} (\Lambda^T)^\nu{}_\beta &= g^{\alpha\beta} \end{aligned}$$

$$\Lambda g \Lambda^T = g$$

2 Properties of the Λ Matrix

$$\det(\Lambda) = \pm 1$$

Taking the determinant of $\Lambda g \Lambda^T = g$:

$$\begin{aligned} \det(\Lambda g \Lambda^T) &= \det(g) \\ \det(\Lambda) \det(g) \det(\Lambda^T) &= \det(g) \\ (\det(\Lambda))^2 &= 1 \end{aligned}$$

$$\det(\Lambda) = \pm 1$$

Boost along the x -axis

$$\Lambda^\mu{}_\nu = \begin{pmatrix} \gamma & -i\beta\gamma & 0 & 0 \\ i\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Check: with $\tan \theta = i\beta = i\frac{v}{c}$,

$$\cos \theta = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \gamma, \quad \sin \theta = i\beta\gamma$$

So a boost can be written formally as a “rotation by an imaginary angle”, matching the ordinary rotation matrix

$$\Lambda^\mu{}_\nu = \begin{pmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (\text{rotation matrix})$$

3 The Lorentz Group

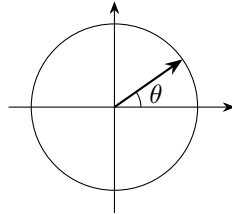
There can be **3 boosts + 3 rotations**, which constitute the **Lorentz group**.

$$SO(3,1) \neq SO(4)$$

$$SO(3,1): \quad -x_0^2 + x_1^2 + x_2^2 + x_3^2 = r^2 \quad \quad SO(4): \quad x_0^2 + x_1^2 + x_2^2 + x_3^2 = r^2$$

Exercise: $SO(2)$

$$R = e^{i\theta\sigma_2}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_2^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$



We expect this to reproduce the familiar 2D rotation

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

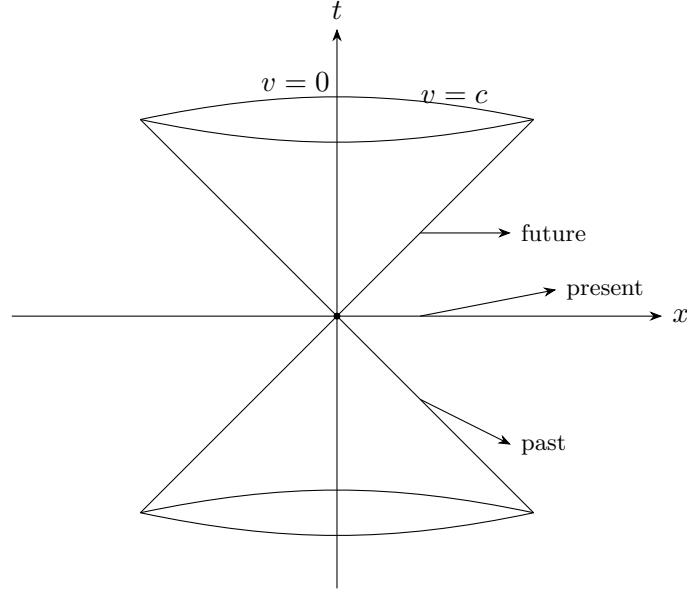
Expand R as a power series:

$$\begin{aligned} R = e^{i\theta\sigma_2} &= 1 + i\theta\sigma_2 - \frac{\theta^2\sigma_2^2}{2!} - \frac{i\theta^3\sigma_2^3}{3!} + \frac{\theta^4\sigma_2^4}{4!} + \dots \\ &= i\sigma_2 \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots \right) + \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots \right) \end{aligned}$$

$$R = e^{i\theta\sigma_2} = i\sigma_2 \sin \theta + \cos \theta \mathbb{I}_{2 \times 2}$$

$\sigma_2 \rightarrow$ generator of this θ -rotation.

4 The Light Cone



5 Worked Example: Spacetime Diagram for a Boost

An inertial system S' moves in the x -direction at a speed $v = \frac{3}{5}c$ relative to S .

- (i) Draw the spacetime diagram representing $x' = -3, -2, -1, 0, 1, 2, 3$.
- (ii) Also draw the lines corresponding to $ct' = -3, -2, -1, 0, 1, 2, 3$.

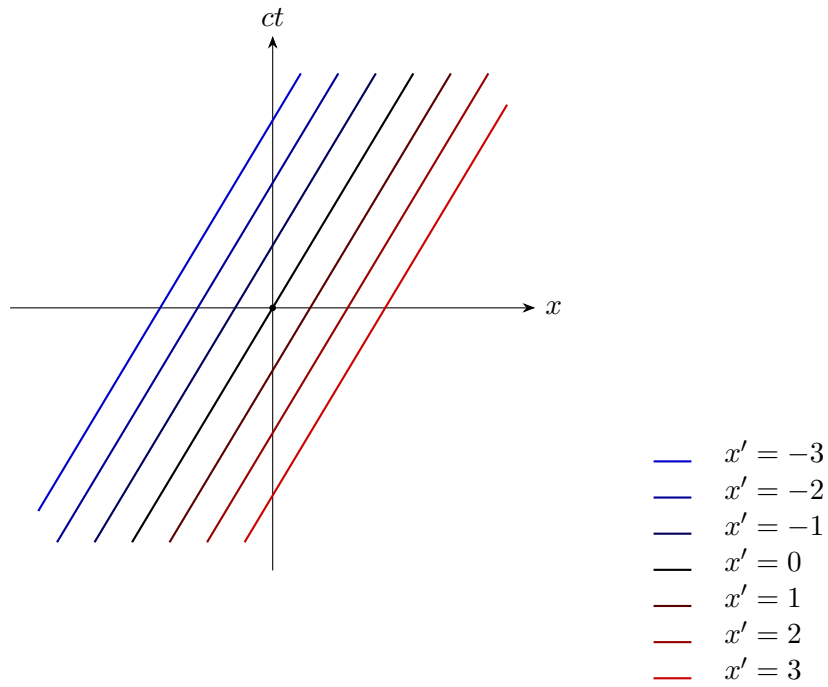
With $\frac{v}{c} = \frac{3}{5}$, $\sqrt{1 - v^2/c^2} = \frac{4}{5}$, $\gamma = \frac{5}{4}$, $\beta = \frac{3}{5}$.

(i) **Lines of constant x'**

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}} = \frac{x - \frac{v}{c} ct}{\frac{4}{5}} = \frac{x - \frac{3}{5} ct}{\frac{4}{5}} = \frac{5x - 3ct}{4}$$

$$x' = \frac{5x - 3ct}{4} \iff ct = \frac{5x - 4x'}{3}$$

for $x' = -3, -2, -1, 0, 1, 2, 3$. Each value of x' traces a straight line of slope $\frac{d(ct)}{dx} = \frac{5}{3}$, parallel to the ct' -axis.



(ii) Lines of constant ct'

$$ct' = \gamma(ct - \beta x) = \frac{5}{4} \left(ct - \frac{3}{5}x \right) = \frac{5}{4}ct - \frac{3}{4}x = \frac{5ct - 3x}{4}$$

$$ct' = \frac{5ct - 3x}{4} \iff ct = \frac{4ct' + 3x}{5}$$

for $ct' = -3, -2, -1, 0, 1, 2, 3$. Each value of ct' traces a straight line of slope $\frac{d(ct)}{dx} = \frac{3}{5}$, parallel to the x' -axis.

